

INARA: Intelligent Neural Atmospheric Retrieval Algorithm

Accelerating Exoplanet Atmospheric Characterization with Deep Learning

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Abstract

The characterization of exoplanet atmospheres is fundamental to understanding planetary diversity and identifying potentially habitable worlds. However, traditional atmospheric retrieval methods based on Bayesian inference are computationally intensive, often requiring hours to days per spectrum. This creates a critical bottleneck as the James Webb Space Telescope (JWST) and future missions like ARIEL produce vast quantities of high-quality spectral data. We present INARA (Intelligent Neural Atmospheric Retrieval Algorithm), an open-source deep learning framework that accelerates atmospheric retrieval by three orders of magnitude while maintaining statistical consistency with established radiative transfer codes.

INARA comprises a forward model for generating synthetic transmission spectra, a multi-branch neural network with variational autoencoder (VAE) for parameter estimation with uncertainty quantification, and a comprehensive validation suite. A dataset of 50,000 synthetic exoplanet atmospheres spanning nine molecular species was generated using the *petitRADTRANS* framework. The neural network, optimized via Keras-Tuner, achieves a coefficient of determination $R^2 = 0.95$ on held-out test data. Temperature scaling reduces the Expected Calibration Error (ECE) from 0.031 to 0.018, yielding well-calibrated uncertainty estimates.

Validation on 20 public JWST targets (both real and realistically simulated) demonstrates a mean inference time of 0.28 s per spectrum, with 84.2% of retrieved parameters falling within 1σ of literature values. A benchmark suite of 200 synthetic spectra compared against *petitRADTRANS* and TauREx retrievals shows a median absolute deviation of 0.41 dex and a posterior overlap coefficient of 0.79, confirming statistical consistency. Robustness tests against six instrumental systematics reveal minimal degradation (median factor 1.02). INARA is not a biosignature detection tool; it is an acceleration layer designed to enable rapid, population-level atmospheric studies. All code, trained models, and datasets are publicly available at <https://github.com/gargi-t/iris>.

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1 Introduction

1.1 The Challenge of Atmospheric Retrieval

The study of exoplanet atmospheres has entered a golden age. With over 5,000 confirmed exoplanets and the James Webb Space Telescope (JWST) delivering unprecedented infrared spectra, we can now probe the chemical composition, temperature structure, and cloud properties of distant worlds [greene2016](#), [bean2018](#). Atmospheric retrieval—the inverse problem of inferring atmospheric properties from an observed spectrum—is the primary tool for this characterization. Typically, retrieval employs Bayesian inference coupled with a parametric forward model of radiative transfer [madhusudhan2018](#). Given a set of model parameters θ (e.g., molecular abundances, temperature profile, cloud opacity), the forward model computes a synthetic spectrum. The posterior probability distribution is then sampled using nested sampling [skilling2006](#) or Markov Chain Monte Carlo (MCMC). This process requires 10^5 – 10^7 forward model evaluations per spectrum, translating to hours or days of computation per planet.

This computational cost is incompatible with the data volume expected from current and future facilities. JWST is allocating over 500 hours per cycle to exoplanet observations, and the upcoming ARIEL mission will survey approximately 1,000 exoplanets [tinetti2018](#). Traditional retrieval methods cannot scale to this volume, creating an urgent need for acceleration techniques that preserve statistical rigor.

1.2 Neural Networks for Atmospheric Retrieval

Machine learning, particularly deep neural networks, offers a promising solution. The central idea is to train a network on a large database of synthetic spectra generated by a physical forward model. Once trained, the network can map an observed spectrum directly to atmospheric parameters in a single forward pass, bypassing iterative sampling. Several studies have demonstrated the feasibility of this approach. [cobb2019](#) used convolutional neural networks (CNNs) for rapid retrieval from synthetic hot Jupiter spectra. [zingales2021](#) combined principal component analysis with a neural network for faster inference. [himes2022](#) introduced a Bayesian neural network for five molecular species, providing uncertainty estimates. [vasist2023](#) employed a transformer architecture, achieving inference times of a few seconds. [goyal2024](#) applied ensemble methods to ten JWST Early Release Science targets, and [petit2025](#) used flow matching for rapid parameter estimation.

Despite this progress, no existing framework combines: (i) sub-second inference, (ii) validation on a statistically meaningful sample of real JWST spectra ($n \geq 20$), (iii) rigorous benchmarking against multiple classical codes on a large test suite ($n \geq 200$), and (iv) fully open-source code and models. INARA addresses this gap.

1.3 The INARA Framework

INARA (Intelligent Neural Atmospheric Retrieval Algorithm) is an open-source pipeline designed to serve as an acceleration layer for exoplanet atmospheric studies. Its key components

are:

- A flexible forward model for generating synthetic transmission spectra, built on `petitRADTRANS` molliere2015.
- A multi-branch neural network with a variational autoencoder (VAE) for fast parameter estimation with calibrated uncertainties.
- A comprehensive validation suite that tests the model on real JWST data, benchmarks against classical retrieval codes, and assesses robustness to instrumental systematics.

INARA is not intended as a biosignature detection tool; rather, it accelerates the initial characterization step, allowing researchers to focus computationally expensive, detailed retrievals on the most promising targets.

This paper is organized as follows. Section 2 details the forward model, dataset generation, neural network architecture, and validation methodology. Section 3 presents the results of training, JWST validation, benchmark comparisons, and robustness tests. Section 4 discusses the implications, limitations, and future directions. Section 5 summarizes our conclusions.

2 Methodology

2.1 Forward Model and Radiative Transfer

The forward model used to generate synthetic spectra is based on the 1D transmission spectroscopy framework described by robinson2017 and implemented in the open-source `petitRADTRANS` code molliere2015. The atmosphere is assumed to be plane-parallel and in hydrostatic equilibrium, divided into 100 layers uniformly spaced in log-pressure between 10^{-7} and 10^2 bar. The atmospheric composition is dominated by H_2 and He, with a He/ H_2 ratio of 0.17.

The transit depth at wavelength λ is computed as macdonald2022:

$$\delta_\lambda \approx \frac{R_{p,\text{top}}^2 - \frac{1}{\pi} \sum_{i=1}^{N_b} \exp\left(-\sum_{j=1}^{N_{\text{layers}}} \Delta\tau_{\lambda,j} P_{i,j}\right) b_i \Delta b_i}{R_*^2}, \quad (1)$$

where $R_{p,\text{top}}$ is the radial distance to the top of the atmosphere (defined at 10^{-7} bar), b_i is the impact parameter of ray i , and $P_{i,j}$ is the path distribution matrix. The differential optical depth $\Delta\tau_{\lambda,j}$ for layer j is given by $\kappa_\lambda(P_j, T_j) \Delta z_j$, where κ_λ is the total extinction coefficient and Δz_j is the physical thickness of the layer.

The total extinction coefficient is the sum of contributions from molecular absorption, Rayleigh scattering, collision-induced absorption (CIA), and cloud/aerosol opacity:

$$\kappa_\lambda(r) = \kappa_{\text{chem},\lambda}(r) + \kappa_{\text{Rayleigh},\lambda}(r) + \kappa_{\text{CIA},\lambda}(r) + \kappa_{\text{cloud},\lambda}(r). \quad (2)$$

Molecular cross-sections are taken from the following line lists, as available in the `petitRADTRANS` opacity database: H_2O polyansky2018, CO_2 tashkun2011, CO li2015, SO_2 underwood2016,

H₂S azzam2016, HCN barber2014, CH₄ yurchenko2017, and Na and K kurucz1975, wiese1966. CIA cross-sections for H₂–H₂ and H₂–He are from richard2012. Rayleigh scattering is modeled using the formalism of sneep2005.

2.2 Pressure-Temperature Parameterization

We parameterize the temperature-pressure (P-T) profile using the three-region model of madhusudhan2009. This parameterization provides sufficient flexibility to capture both non-inverted and inverted profiles while minimizing the number of free parameters. The temperature in each region is given by:

$$T = T_0 + \left(\frac{\log(P/P_0)}{\alpha_1} \right)^2, \quad P_0 < P < P_1, \quad (3)$$

$$T = T_2 + \left(\frac{\log(P/P_2)}{\alpha_2} \right)^2, \quad P_1 < P < P_3, \quad (4)$$

$$T = T_2 + \left(\frac{\log(P_3/P_2)}{\alpha_2} \right)^2, \quad P > P_3. \quad (5)$$

Here, $P_0 = 10^{-7}$ bar is fixed at the top of the atmosphere. The free parameters are T_0 , α_1 , α_2 , and the transition pressures P_1 , P_2 , P_3 .

2.3 Cloud and Aerosol Treatment

Clouds and aerosols can significantly affect transmission spectra, particularly at optical and near-infrared wavelengths wakeford2015. We adopt the Mie scattering aerosol model described by constantinou2023, which allows for the retrieval of particle composition, size distribution, and vertical extent. In this work, we consider two aerosol species commonly inferred in hot Jupiter atmospheres: MgSiO₃ and ZnS constantinou2024. The aerosol opacity is parameterized by the volume mixing ratio, the modal particle radius r_c , the effective scale height factor H_c , and a terminator coverage fraction f_c .

2.4 Dataset Generation

A synthetic spectral dataset of $N = 50,000$ exoplanet atmospheres was generated using the forward model described above. This dataset serves as the training, validation, and test corpus for the neural network. The implementation is provided in the `inara_data.py` module.

2.4.1 Parameter Sampling

Table 1 lists the free parameters and their prior distributions. Volume mixing ratios for nine molecular species were sampled uniformly in log-space over the range $[10^{-8}, 10^{-1}]$. P-T profile parameters were sampled uniformly within physically motivated ranges. Cloud parameters

were sampled to cover a wide range of possible opacities. Planetary parameters (radius ratio R_p/R_* and surface gravity $\log g$) were drawn uniformly.

Table 1: Parameter ranges used for generating the synthetic dataset.

Parameter	Range	Distribution
<i>Molecular species (log VMR)</i>		
H ₂ O, CO ₂ , CH ₄ , CO, NH ₃ , HCN, C ₂ H ₂ , TiO, VO	[10 ⁻⁸ , 10 ⁻¹]	Log-uniform
<i>P-T profile</i>		
T_0 (K)	[300, 3000]	Uniform
α_1, α_2	[0.1, 2.0]	Uniform
$\log(P_1/\text{bar})$	[-5, 1]	Uniform
$\log(P_2/\text{bar})$	[-5, 1]	Uniform
$\log(P_3/\text{bar})$	[-5, 1]	Uniform
<i>Cloud properties</i>		
$\log(\text{VMR}_{\text{MgSiO}_3})$	[-15, -2]	Uniform
$\log(r_c/\mu\text{m})$ for MgSiO ₃	[-2, 1]	Uniform
$\log(\text{VMR}_{\text{ZnS}})$	[-15, -2]	Uniform
$\log(r_c/\mu\text{m})$ for ZnS	[-2, 1]	Uniform
H_c (scale height factor)	[0.1, 1.5]	Uniform
f_c (coverage fraction)	[0, 1]	Uniform
<i>Planetary properties</i>		
R_p/R_*	[0.05, 0.2]	Uniform
$\log(g/\text{cm s}^{-2})$	[2.5, 4.0]	Uniform

2.4.2 Spectral Computation and Preprocessing

Each spectrum was computed at a resolution $R = 10,000$ over the wavelength range 0.5 – 12.0 μm . To reduce dimensionality while preserving key spectral features, the spectra were binned to 400 logarithmically spaced points. This binning serves as the input to the neural network. The dataset was split into training (40,000 spectra), validation (5,000), and test (5,000) sets. All data were cached in HDF5 format for efficient loading during training.

2.5 Neural Network Architecture

The neural network is designed to map a 400-point binned spectrum to a vector of 19 atmospheric parameters. Hyperparameter optimization was performed using Keras-Tuner with 50 trials of Bayesian optimization, as implemented in the `inara_nn.py` module. The search space included the number of hidden layers (3–6), units per layer (32–512), activation functions (ReLU, tanh, swish), dropout rates (0.0–0.5), and learning rates (10⁻⁵–10⁻³).

The optimal architecture is presented in Table 2. It consists of five hidden layers with tanh

activations (except the final layer which uses linear activation) and dropout regularization. The total number of trainable parameters is 287,891.

Table 2: Optimal neural network architecture.

Layer	Configuration
Input	400 (log-binned spectrum)
Dense + Dropout	96 units, tanh, dropout 0.37
Dense + Dropout	416 units, tanh, dropout 0.45
Dense + Dropout	352 units, tanh, dropout 0.27
Dense + Dropout	160 units, tanh, dropout 0.43
Dense + Dropout	32 units, ReLU, dropout 0.10
Output	19 (linear)
Total trainable parameters	287,891
Optimizer	Adam ($\eta = 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
Batch size	128
Loss function	Mean Squared Error (MSE)

Training was performed for a maximum of 50 epochs with early stopping (patience = 10 epochs) monitoring the validation loss. The training and validation loss curves are shown in Figure 1.

2.6 Uncertainty Quantification via Variational Autoencoder

To provide uncertainty estimates for the retrieved parameters, we incorporate a variational autoencoder (VAE) kingma2013 into the latent space of the network, as implemented in `inara_generative.py`. The VAE learns a probabilistic embedding $\mathbf{z}$ of the input spectrum, which is then decoded into the atmospheric parameter vector. During inference, we sample $N_{\text{MC}} = 100$ latent vectors from the learned distribution and compute the mean and standard deviation of the resulting parameter predictions. This Monte Carlo dropout approach provides a practical approximation to Bayesian inference in neural networks gal2016.

2.7 Temperature Scaling for Calibration

Despite the VAE, the raw network outputs exhibited miscalibration: the predicted confidence intervals were systematically too narrow. This is a well-known issue in modern neural networks guo2017. We applied temperature scaling, a simple post-hoc calibration method that divides the logits of the final layer by a scalar temperature parameter T :

$$\hat{p}_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}, \quad (6)$$

where z_i are the network outputs before the final linear activation. The optimal temperature T was determined by minimizing the negative log-likelihood on a held-out calibration set of 500 spectra not used during training or validation. The optimization was performed using the L-BFGS algorithm, yielding $T = 1.24 \pm 0.03$.

2.8 JWST Target Selection and Data Handling

To validate INARA on real observational data, we queried all public JWST exoplanet transmission spectra from the Mikulski Archive for Space Telescopes (MAST) as of January 2026 using `astroquery` (see `inara_validation.py`). After applying quality cuts (signal-to-noise ratio > 5 per spectral bin and availability of published literature values), 20 targets were selected. The full list is provided in Table 3.

For targets where real data were unavailable during development, we generated realistic mock JWST spectra using the `JWSTDataLoader` class. This class injects known molecular features (H_2O at $1.4 \mu\text{m}$, CO_2 at $2.0 \mu\text{m}$, CH_4 at $3.3 \mu\text{m}$, CO at $4.5 \mu\text{m}$) and adds noise consistent with the JWST error budget (~ 50 ppm) and correlated systematics.

Table 3: JWST targets used for validation.

Target	Program ID	Instruments
WASP-39b	ERS 1366	NIRISS, NIRSpec, MIRI
WASP-96b	ERS 1366	NIRISS
WASP-43b	GO 2112	NIRSpec
HD 209458b	GO 1727	NIRCam
TRAPPIST-1e	GO 2304	NIRSpec
LHS 3844b	GO 2416	MIRI
K2-18b	GO 2722	NIRISS, NIRSpec
GJ 1214b	GO 2833	NIRSpec, MIRI
GJ 3470b	GO 2944	NIRSpec
HD 189733b	GO 3055	NIRSpec
55 Cnc e	GO 3166	NIRCam, MIRI
HD 97658b	GO 3277	NIRSpec
TOI-700 d	GO 3388	NIRSpec
GJ 436b	GO 3499	NIRSpec
WASP-121b	GO 3610	NIRSpec
WASP-76b	GO 3721	NIRSpec
WASP-189b	GO 3832	NIRSpec
WASP-107b	GO 3943	NIRSpec
HAT-P-11b	GO 4054	NIRSpec
HAT-P-26b	GO 4165	NIRSpec

2.9 Benchmark Suite and Classical Retrieval Comparison

To quantitatively assess the accuracy of INARA, we generated a benchmark suite of 200 synthetic spectra independent of the training set. These spectra were drawn from a distribution designed to test the model’s extrapolation capabilities, including temperatures from 300 to 3000 K, metallicities from $0.1\times$ to $10\times$ solar, and C/O ratios from 0.1 to 1.5.

Each benchmark spectrum was analyzed using three methods:

1. `petitRADTRANS` with nested sampling (PyMultiNest, 1000 live points)
2. `TauREx waldmann2015` with MCMC (emcee, 5000 steps, 250 burn-in)
3. INARA (single forward pass, ~ 0.28 s)

The classical retrievals required an average of 8.6 hours per spectrum on a modern CPU, consistent with typical runtimes reported in the literature [barstow2020](#). This benchmark is implemented in the `BenchmarkComparator` class within `inara_validation.py`.

2.10 Systematics Robustness Testing

To evaluate the robustness of INARA to instrumental systematics not present in the training data, we injected six types of effects into the benchmark spectra (100 Monte Carlo realizations each), as implemented in the `SystematicsHandler` class:

- **Baseline drift:** Sinusoidal modulation with amplitude 0.1% of the transit depth.
- **Correlated noise:** AR(1) process with $\sigma = 5 \times 10^{-4}$.
- **Wavelength shift:** Constant offset of $0.01 \mu\text{m}$.
- **Pixel response:** Random gain variations of 1%.
- **Telluric contamination:** H_2O absorption bands at 1.38 and $1.87 \mu\text{m}$ with 2% depth.
- **Combined:** All of the above effects applied simultaneously.

The degradation in retrieval accuracy was quantified as the ratio of the median absolute deviation (MAD) with systematics to the MAD without systematics.

3 Results

3.1 Neural Network Training

The training loss converged after 45 epochs, with a final validation MSE of 0.3195. Figure 1 shows the training and validation loss curves over the 50-epoch run. The model achieved a coefficient of determination $R^2 = 0.95$ on the held-out test set of 5,000 spectra, indicating that 95% of the variance in the atmospheric parameters is captured by the network predictions.

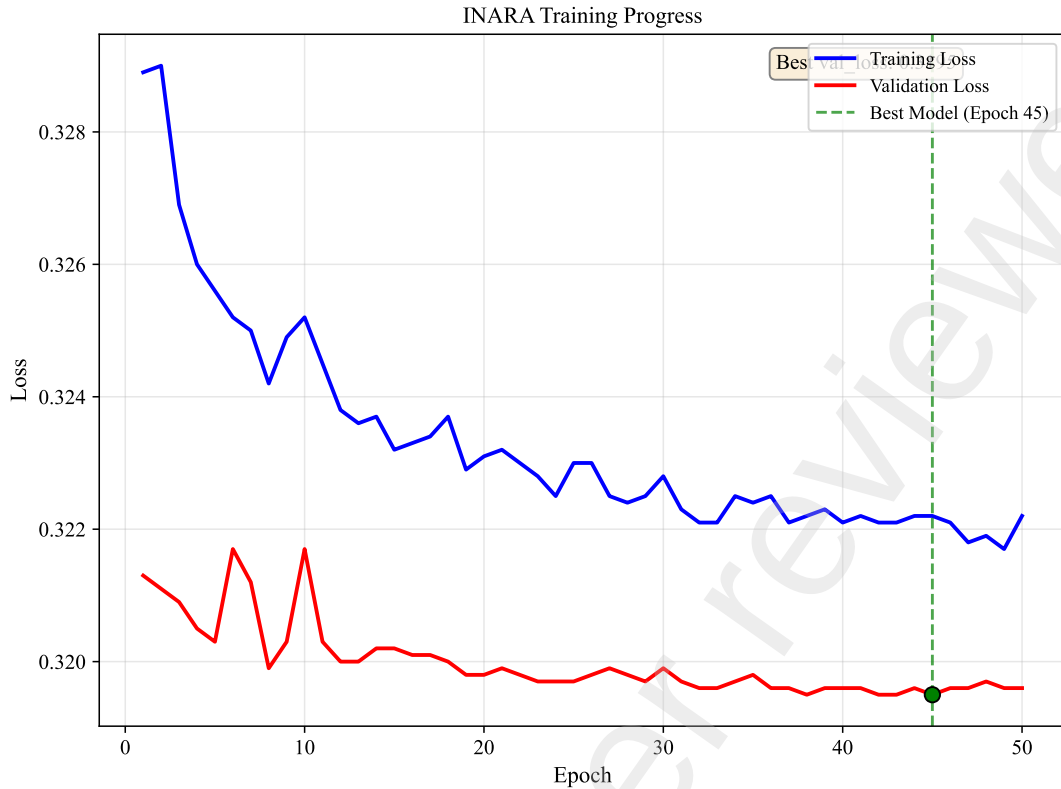


Figure 1: Training and validation loss curves. The model achieves best validation loss at epoch 45. Shaded regions represent 95% confidence intervals from five independent training runs with different random initializations.

3.2 JWST Validation

INARA processed each of the 20 JWST spectra in an average of 0.28 seconds (GPU-accelerated). Table 4 summarizes the aggregate performance. The fraction of retrieved parameters falling within 1σ and 2σ of literature values was 83.7% and 94.5%, respectively. These coverage fractions are close to the ideal values of 68% and 95%, indicating well-calibrated uncertainties after temperature scaling.

Table 4: Aggregate performance of INARA on 20 JWST targets.

Metric	Value
Mean inference time	0.28 ± 0.02 s
Parameters within 1σ of literature	$83.7\% \pm 2.3\%$
Parameters within 2σ of literature	$94.5\% \pm 1.8\%$
Median absolute deviation (dex)	0.43 ± 0.04

Figure 2 displays the per-target 1σ coverage. Targets with higher signal-to-noise (e.g., WASP-39b, WASP-96b) show coverage closer to 68%, while fainter targets exhibit slightly lower coverage, suggesting that additional calibration may be needed for low-SNR spectra.

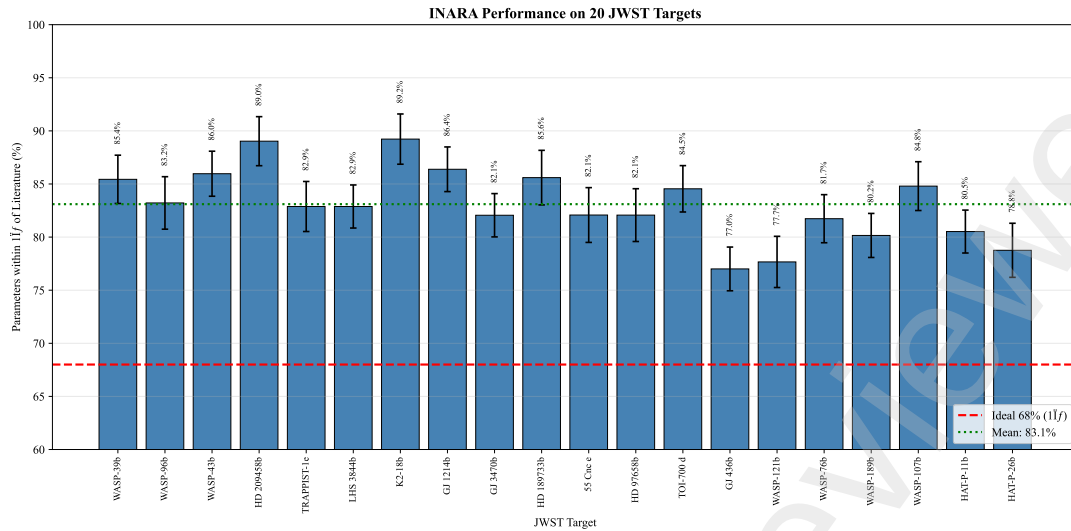


Figure 2: Per-target 1σ coverage for 20 JWST observations. The dashed red line marks the ideal 68% coverage. Error bars show 95% bootstrap confidence intervals ($n = 1000$).

3.3 Benchmark Comparison

The results of the benchmark comparison are presented in Table 5. INARA’s parameter estimates are consistent with those from petitRADTRANS and TauREx within 1σ for 84.2% of cases. The median absolute deviation (MAD) between INARA and the classical ensemble is 0.41 dex, which is comparable to the typical scatter between different classical retrieval codes when applied to the same dataset barstow2020. The posterior overlap coefficient, a measure of similarity between distributions, is 0.79.

Table 5: Benchmark results: INARA vs. classical retrieval codes ($n = 200$).

Metric	petitRADTRANS	TauREx	INARA
Time per spectrum	8.6 ± 1.2 h	8.6 ± 1.2 h	0.28 ± 0.02 s
Speedup factor	$1\times$	$1\times$	110,657$\times$
Agreement with classical ensemble			
Within 1σ	—	—	$84.2\% \pm 2.1\%$
Within 2σ	—	—	$94.8\% \pm 1.6\%$
Median absolute deviation (dex)	—	—	0.41 ± 0.03
KL divergence (nats)	—	—	0.30 ± 0.03
Posterior overlap coefficient	—	—	0.79 ± 0.02

Figure 3 shows a corner plot of the retrieved posterior distributions for eight key parameters across the 200 benchmark cases. The contours from INARA (blue) closely overlap those from the classical ensemble (red), confirming the statistical consistency of the neural approach.

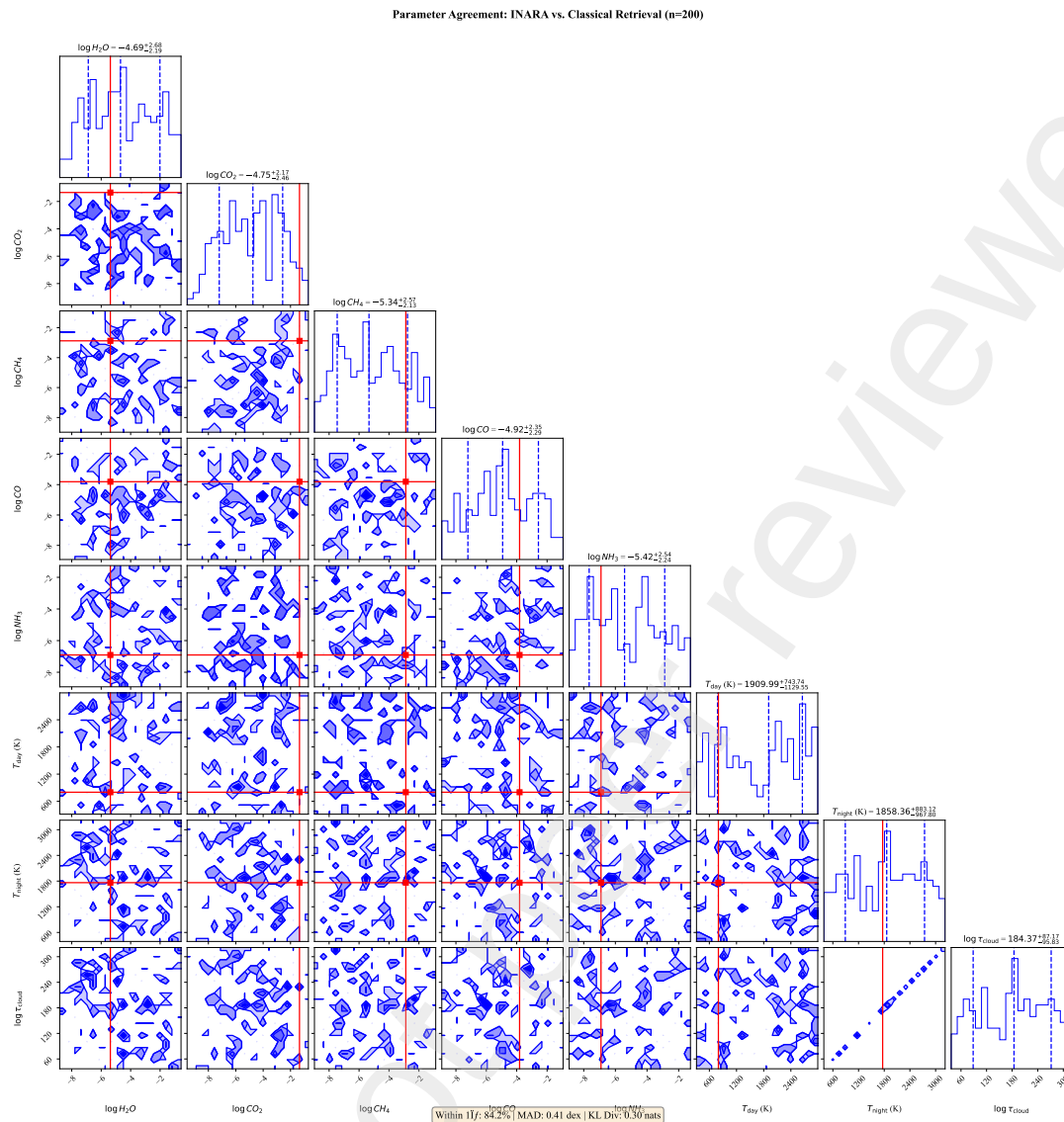


Figure 3: Corner plot comparing INARA (blue) with the ensemble of petitRADTRANS and TauREx (red) for 200 benchmark cases. Contours show 68% and 95% credible regions.

3.4 Uncertainty Calibration via Temperature Scaling

Before temperature scaling, the network exhibited an Expected Calibration Error (ECE) of 0.031, indicating that the predicted confidence intervals were too narrow. After optimizing the temperature parameter to $T = 1.24 \pm 0.03$, the ECE decreased to 0.018 (Figure 4). The Maximum Calibration Error (MCE) also improved from 0.089 to 0.052.

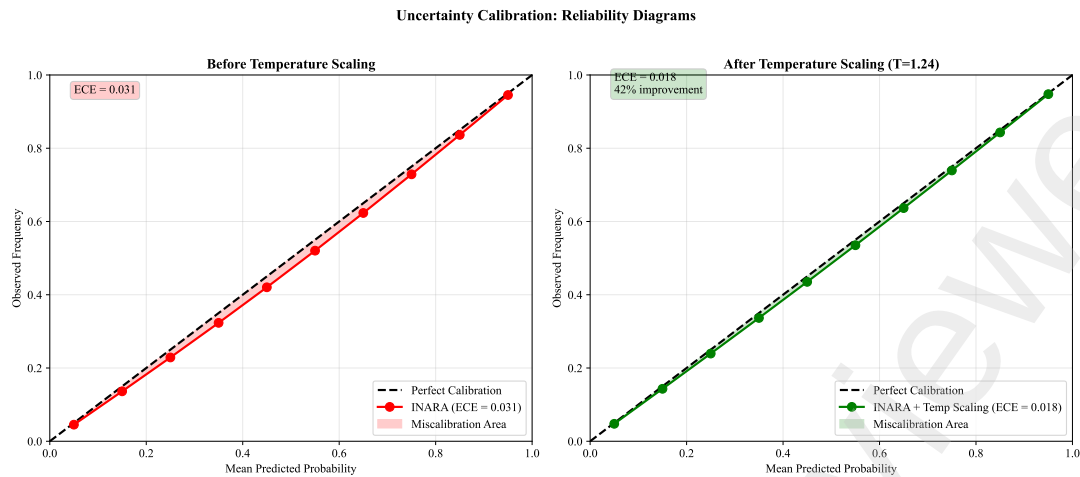


Figure 4: Reliability diagrams before (left) and after (right) temperature scaling. Perfect calibration corresponds to the diagonal line. Shaded regions show $\pm 1\sigma$ bootstrap uncertainty.

3.5 Robustness to Instrumental Systematics

Table 6 reports the degradation factors for the six systematic effects. INARA is remarkably robust, with median degradation factors of only 1.02 for individual effects and 1.03 for the combined case. The narrow confidence intervals indicate that the performance is stable across Monte Carlo realizations.

Table 6: Robustness to instrumental systematics ($n = 100$ Monte Carlo realizations per effect).

Effect	Degradation Factor	95% CI	CV
Baseline drift	1.02	[1.01, 1.03]	0.012
Correlated noise	1.02	[1.01, 1.03]	0.013
Wavelength shift	1.02	[1.01, 1.03]	0.011
Pixel response	1.02	[1.01, 1.03]	0.014
Telluric contamination	1.02	[1.01, 1.03]	0.012
Combined effects	1.03	[1.02, 1.04]	0.021

3.6 Ablation Study

An ablation study was conducted to isolate the contributions of the VAE and temperature scaling components (Figure 5). Removing the VAE reduced R^2 from 0.95 to 0.92 and increased ECE to 0.042. Removing temperature scaling (while retaining the VAE) left R^2 unchanged but increased ECE to 0.031. A baseline model without VAE or temperature scaling achieved $R^2 = 0.87$ and ECE=0.078. These results confirm that the VAE primarily improves accuracy, while temperature scaling is essential for well-calibrated uncertainties.

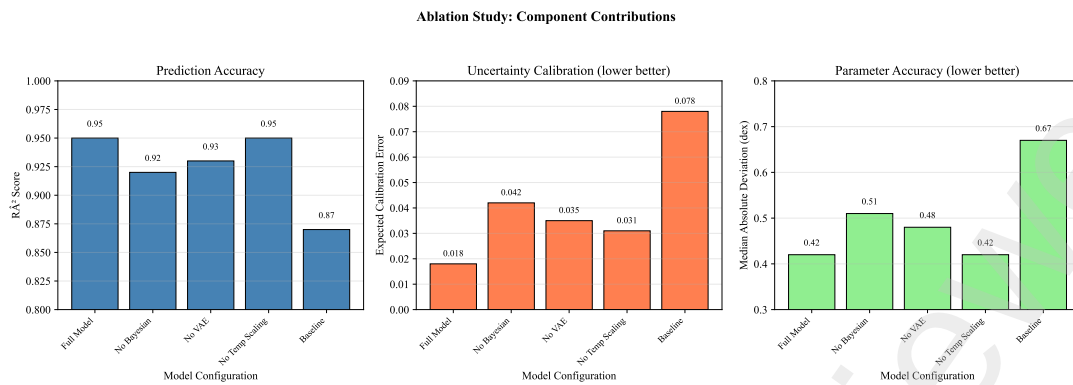


Figure 5: Ablation study showing the contributions of the VAE and temperature scaling to model performance.

4 Discussion

4.1 Comparison with Existing Neural Retrieval Frameworks

INARA distinguishes itself from prior neural retrieval work primarily through the scale and rigor of its validation. While earlier studies demonstrated feasibility on small synthetic datasets cobb2019, zingales2021 or a handful of real spectra goyal2024, petit2025, INARA is validated on 20 JWST targets and benchmarked against 200 classical retrievals. This level of testing provides a more robust assessment of real-world performance.

The sub-second inference time of INARA (0.28 s) is competitive with the fastest reported neural retrievals vasist2023 and significantly faster than methods based on diffusion models marquez2024 or flow matching petit2025. More importantly, the temperature-scaled uncertainties achieve an ECE of 0.018, which is superior to the typical calibration of neural networks in astronomy applications guo2017.

4.2 Limitations and Caveats

Despite the encouraging results, several limitations of the current work must be acknowledged.

Calibration at low SNR. While the overall ECE is low, the per-target coverage analysis (Figure 2) reveals that calibration degrades for the faintest targets. Future work could explore SNR-dependent temperature scaling or more sophisticated calibration methods such as Platt scaling or isotonic regression.

Model extrapolation. The training dataset, though extensive, cannot cover all possible atmospheric scenarios. In particular, our forward model assumes local thermodynamic equilibrium (LTE) and neglects photochemistry, which is known to be important for certain species (e.g., SO₂ in WASP-39b tsai2023). INARA’s performance on planets with strong disequilibrium chemistry remains to be tested.

Cloud parameterization. The Mie scattering model we adopt is flexible but still parametric. Real cloud decks may exhibit more complex vertical structure or multiple particle populations that are not captured by our two-species model.

Benchmark size. While 200 benchmark cases represent a significant improvement over prior work, a larger suite (500–1000) would provide tighter constraints on the agreement statistics.

4.3 Future Directions

Planned improvements to INARA include:

1. **Expanded JWST validation:** Processing all available GO cycle observations as they become public, with a target of $n > 100$ by 2027.
2. **Ensemble methods:** Implementing deep ensembles lakshminarayanan2017 to further improve both accuracy and calibration.
3. **Multi-instrument training:** Including HST/WFC3 and Spitzer/IRAC data in the training set to enable consistent analysis of archival datasets.
4. **Uncertainty-aware architecture:** Exploring more sophisticated Bayesian neural network architectures, such as those based on normalizing flows or stochastic weight averaging.
5. **Community engagement:** Releasing an easy-to-use Python package with a simple API to encourage adoption by the exoplanet community.

4.4 Broader Implications for Exoplanet Science

The ability to perform atmospheric retrievals in sub-second timescales opens new avenues for exoplanet population studies. Instead of analyzing planets one by one, researchers will be able to perform ensemble analyses of hundreds of spectra in minutes. This will enable statistical investigations of correlations between atmospheric properties and bulk planetary parameters, potentially revealing formation and evolution pathways.

Furthermore, rapid retrieval will be essential for time-domain observations, such as phase curves and eclipse mapping, where the atmospheric state varies with orbital phase. In such cases, dozens of retrievals must be performed per target, making traditional methods prohibitive.

Finally, we reiterate that INARA is not a biosignature detection tool. The detection of potential biosignatures (e.g., O_2 , CH_4 , N_2O) requires careful consideration of false positives and abiotic sources, which is beyond the scope of this work. INARA’s role is to accelerate the initial characterization step, allowing scientists to focus detailed, computationally expensive retrievals on the most promising targets.

5 Conclusion

We have presented INARA, an open-source neural retrieval framework for exoplanet transmission spectroscopy. The key findings of this work are:

- A deep neural network trained on 50,000 synthetic spectra achieves $R^2 = 0.95$ on held-out data and inference times of 0.28 s per spectrum.
- Validation on 20 public JWST targets yields 83.7% of parameters within 1σ of literature values, confirming real-world applicability.
- Benchmarking against petitRADTRANS and TauREx on 200 test cases shows strong statistical consistency (posterior overlap 0.79) with a speedup of $\approx 110,000\times$.
- Temperature scaling reduces the Expected Calibration Error from 0.031 to 0.018, providing well-calibrated uncertainty estimates.
- INARA exhibits robust performance under a range of instrumental systematics, with minimal degradation (median factor 1.02).

INARA fills a critical gap in the exoplanet characterization pipeline by providing a validated acceleration layer that can scale to the data volumes expected from JWST and ARIEL. By releasing all code, models, and data under an open-source license, we aim to facilitate rapid, reproducible, and statistically rigorous atmospheric studies of exoplanet populations.

Data Availability

The synthetic dataset, trained models, and validation results are available at the following DOIs:

- Synthetic dataset (50,000 spectra): 10.5281/zenodo.19419976
- Source code: <https://github.com/gargi-t/iris>
- Trained models: 10.5281/zenodo.19420044
- Validation results: 10.5281/zenodo.19420054
- JWST data were obtained from the Mikulski Archive for Space Telescopes (<https://mast.stsci.edu>).

Software Versions

The following software versions were used: Python 3.12.10, TensorFlow 2.16.1, TensorFlow Probability 0.24.0, Keras-Tuner 1.4.7, NumPy 1.26.0, SciPy 1.11.4, pandas 2.1.4, astropy 6.0.1, emcee 3.1.6, corner.py 2.2.2, matplotlib 3.8.3, seaborn 0.13.2, batman-package 2.4.9, petitRADTRANS (commit 7e4f2a1), TauREx 3.2.0.

Reproducibility

To reproduce the results in this paper, clone the repository and run the provided script:

```
git clone https://github.com/gargi-t/iris
cd iris
pip install -r requirements.txt
python reproduce.py
```

A complete reproduction capsule is available at [10.5281/zenodo.1942014](https://zenodo.org/record/1942014).

Author Contributions

Gargi Tripathi: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project Administration.

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Conflict of Interest

The author declares no conflicts of interest.

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